

■ Docker Security Audit Lab

Project 06 — Complete Walkthrough

This document captures all steps of the Docker Security Audit lab, from project setup through container networking, volume management, service connectivity, and audit report generation.

Lab Steps Covered:

- Step 1: Project Directory Created — mkdir reports, tree shows structure
- Step 2: Docker Compose YML Skeleton — services: nginx, postgres, audit-tool
- Step 3: Docker Compose File — nginx, postgres config with env_file, volumes, ports
- Step 4: Custom Bridge Network Created — docker network create security-net
- Step 5: Named Volumes Created — nginx-logs, postgres-data, audit-data
- Step 6: Volumes Verified — docker volume ls confirms all 3 volumes
- Step 7: Containers Up — docker compose up -d → postgres, nginx, audit-tool all started
- Step 8: Network Inspect — security-net shows all 3 containers with IPs
- Step 9: Volume Inspect — mountpoints verified for all volumes
- Step 10: Install iputils-ping — apt install inside audit-tool container
- Step 11: Ping Test — audit-tool pings nginx (172.18.0.3) and postgres (172.18.0.4) successfully
- Step 12: curl http://nginx — audit-tool reaches nginx container, returns full HTML
- Step 13: localhost:8080 — nginx accessible from Windows browser via port mapping
- Step 14: Install PostgreSQL Client — apt install postgresql-client inside audit-tool
- Step 15: psql Connection — psql -h postgres -U postgres connects successfully
- Step 16: PostgreSQL \l — List of databases: companydb, postgres, template0, template1
- Step 17: Audit Report Output — docker-audit-2026-06-17.txt shows images, networks, volumes, stats, ports
- Step 18: Final Project Tree — README.md, docker-compose.yml, audit.sh, postgres.env, reports/audit.txt

Step 1: Project Directory Created — mkdir reports, tree shows structure

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS bash - project-06-docker-security-aud

syedaftab04@SYEDLENOVO:~/docker-labs$ ls
● syedaftab04@SYEDLENOVO:~/docker-labs$ cd project-06-docker-security-audit
● syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ ls
● syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ mkdir reports
● syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ tree
Command 'tree' not found, but can be installed with:
sudo snap install tree # version 2.1.3+pkg-5852, or
sudo apt install tree # version 2.3.1-1
See 'snap info tree' for additional versions.
● syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ sudo apt install tree
[sudo: authenticate] Password:
Installing:
tree

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
  Download size: 53.5 kB
  Space needed: 125 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 tree amd64 2.3.1-1 [53.5 kB]
Fetched 53.5 kB in 0s (940 kB/s)
Selecting previously unselected package tree.
(Reading database ... 35927 files and directories currently installed.)
Preparing to unpack .../tree_2.3.1-1_amd64.deb ...
Unpacking tree (2.3.1-1) ...
Setting up tree (2.3.1-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
● syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ tree
.
├── reports

2 directories, 0 files
```

Step 2: Docker Compose YML Skeleton — services: nginx, postgres, audit-tool

```
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ cat docker-compose.yml
services:

  nginx:

  postgres:

  audit-tool:

networks:

  security-net:
    external: true

volumes:

  postgres-data:
    external: true

  nginx-logs:
    external: true

  audit-data:
    external: true
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$
```



Step 3: Docker Compose File — nginx, postgres config with env_file, volumes, ports

```
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ cat docker-compose.yml
```

```
nginx:
  image: nginx
  container_name: nginx

  networks:
    - security-net

  volumes:
    - nginx-logs:/var/log/nginx

  ports:
    - "8080:80"

postgres:

  image: postgres
  container_name: postgres

  env_file:
    - postgres.env

  ports:
    - "5432:5432"

  networks:
    - security-net

  volumes:
    - postgres-data:/var/lib/postgresql
```

0 0 0



Step 4: Custom Bridge Network Created — docker network create security-net

StartWalkthroughs

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

bash - project-06-docker-security-audit + -

```
syedaftab04@SYEDLENOVO:~/docker-labs/Multi-stage-dockerfile$ cd ..
● syedaftab04@SYEDLENOVO:~/docker-labs$ ls
'# Docker Volumes Lab '$'\342\200\223'' Data Persistence with Bind Mounts' project-04-container-resource-monitoring
Multi-stage-dockerfile project-04-multistage-resource-monitoring
README.md project-05-failed-container-investigation
java-docker-app project-06-docker-security-audit
nodejs-docker-app shared-data
project-01-python-flask-docker
● syedaftab04@SYEDLENOVO:~/docker-labs$ cd project-06-docker-security-audit
● syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ ls
README.md docker-compose.yml docker-security-audit.sh postgres.env reports
● syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ tree
.
├── README.md
├── docker-compose.yml
├── docker-security-audit.sh
├── postgres.env
└── reports

2 directories, 4 files
● syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker network create security-net
b3cce70deccfca517bd0db1784ee6b63b5d4562bf11f125a13863c155422218b
● syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
b2d38f3b7c3c        bridge              bridge              local
383de4df043e        host                host                local
7772ea80cf7b        none                null                local
b3cce70deccf        security-net        bridge              local
○ syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$
```

Step 5: Named Volumes Created — nginx-logs, postgres-data, audit-data

```
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker volume inspect postgres-data
[
  {
    "CreatedAt": "2026-06-17T13:43:18Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/postgres-data/_data",
    "Name": "postgres-data",
    "Options": null,
    "Scope": "local"
  }
]
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker volume inspect nginx-logs
[
  {
    "CreatedAt": "2026-06-17T13:43:18Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/nginx-logs/_data",
    "Name": "nginx-logs",
    "Options": null,
    "Scope": "local"
  }
]
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker volume inspect audit-data
[
  {
    "CreatedAt": "2026-06-17T13:43:29Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/audit-data/_data",
    "Name": "audit-data",
    "Options": null,
    "Scope": "local"
  }
]
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$
```


Step 6: Volumes Verified — docker volume ls confirms all 3 volumes

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS bash - project-06-docker-security-audit + v [ ] [ ]

syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ ls
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ tree
.
├── README.md
├── docker-compose.yml
├── docker-security-audit.sh
├── postgres.env
└── reports

2 directories, 4 files
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker network create security-net
b3cce70deccfca517bd0db1784ee6b63b5d4562bf11f125a13863c155422218b
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
b2d38f3b7c3c        bridge              bridge              local
383de4df043e        host                host                local
7772ea80cf7b        none                null                local
b3cce70deccf        security-net        bridge              local
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker volume create nginx-logs
nginx-logs
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker volume create postgres-data
postgres-data
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker volume create audit-data
audit-data
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker volume inspect ls
[]
Error response from daemon: get ls: no such volume
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker volume ls
DRIVER              VOLUME NAME
local                audit-data
local                nginx-logs
local                postgres-data
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$
```



Step 7: Containers Up — docker compose up -d → postgres, nginx, audit-tool all started

```
name: postgres-data
external: true

syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker compose up -d
[+] up 18/18
✓ Image postgres Pulled 12.3s
✓ Container postgres Started 1.0s
✓ Container nginx Started 1.0s
✓ Container audit-tool Started 0.9s

syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
4d67cdaa70f1   postgres      "docker-entrypoint.s..." 30 seconds ago Up 29 seconds 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp postgres
36d7f0c7b877   nginx         "/docker-entrypoint..." 30 seconds ago Up 29 seconds 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp nginx
9eab4d718a89   ubuntu       "sleep infinity"          30 seconds ago Up 29 seconds                          audit-too
1

syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$
```


Step 8: Network Inspect — security-net shows all 3 containers with IPs

```
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker network inspect security-net
[
  {
    "Labels": {},
    "Containers": {
      "36df70c7b87737adee7324fe3c723760498a5574ef73adbb9caf8cf11edee696": {
        "Name": "nginx",
        "EndpointID": "9b9c528d84b8d83448e2b3fbb27561ffdf32711596ab2ef1859059c973028699",
        "MacAddress": "ee:66:5c:13:3d:0e",
        "IPv4Address": "172.18.0.3/16",
        "IPv6Address": ""
      },
      "4d67cdaa70f1d525a02503b02a554d37b273f08f8318e2ae2809ea2a43910f14": {
        "Name": "postgres",
        "EndpointID": "c436964249088aea48e0052c600e35aca0fd67ad134b10ce916887f4f98137ab",
        "MacAddress": "4a:83:0b:1e:91:ec",
        "IPv4Address": "172.18.0.4/16",
        "IPv6Address": ""
      },
      "9eab4d718a89a2e17ec58f4ab5b54e463a3c0c96d8f90fc6dba9ca442a03ea2c": {
        "Name": "audit-tool",
        "EndpointID": "eb26a6276d07da641c440a2e1ddf20b4fb7dc4129f2ff94c471b2a66a70fc5d6",
        "MacAddress": "66:26:89:33:67:99",
        "IPv4Address": "172.18.0.2/16",
        "IPv6Address": ""
      }
    },
    "Status": {
      "IPAM": {
        "Subnets": {
          "172.18.0.0/16": {
            "IPsInUse": 6,
            "DynamicIPsAvailable": 65530
          }
        }
      }
    }
  }
]
```

Step 9: Volume Inspect — mountpoints verified for all volumes

```
]
● syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker volume ls
DRIVER      VOLUME NAME
local       audit-data
local       nginx-logs
local       postgres-data
○ syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$
```

The image shows a Windows taskbar at the bottom of the terminal window. It includes a search bar on the left, followed by several application icons: a cloud storage icon, a file explorer icon, a Microsoft Edge icon, a file icon, a Teams icon with a green checkmark, a Visual Studio Code icon, a Google Chrome icon, a calendar icon, and a Docker icon.

Step 10: Install iputils-ping — apt install inside audit-tool container

```
local postgres-data
● syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker exec -it audit-tool bash
root@9eab4d718a89:/# apt update
apt install iputils-ping -y
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
```

Step 11: Ping Test — audit-tool pings nginx (172.18.0.3) and postgres (172.18.0.4) successfully

```
64 bytes from nginx.security-net (172.18.0.3): icmp_seq=3 ttl=64 time=0.112 ms
64 bytes from nginx.security-net (172.18.0.3): icmp_seq=4 ttl=64 time=0.084 ms
^C
--- nginx ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3040ms
rtt min/avg/max/mdev = 0.083/0.395/1.302/0.523 ms
root@9eab4d718a89:/# ping postgres
PING postgres (172.18.0.4) 56(84) bytes of data.
64 bytes from postgres.security-net (172.18.0.4): icmp_seq=1 ttl=64 time=0.203 ms
64 bytes from postgres.security-net (172.18.0.4): icmp_seq=2 ttl=64 time=0.084 ms
64 bytes from postgres.security-net (172.18.0.4): icmp_seq=3 ttl=64 time=0.102 ms
64 bytes from postgres.security-net (172.18.0.4): icmp_seq=4 ttl=64 time=0.097 ms
^C
--- postgres ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3054ms
rtt min/avg/max/mdev = 0.084/0.121/0.203/0.047 ms
root@9eab4d718a89:/#
```

0 0 0 0



Search



Step 12: curl http://nginx — audit-tool reaches nginx container, returns full HTML

```
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker exec -it audit-tool bash
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
root@9eab4d718a89:/# curl http://nginx
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
root@9eab4d718a89:/#
```

0 0



Step 13: localhost:8080 — nginx accessible from Windows browser via port mapping

← → ↻ ⓘ localhost:8080

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.

To engage with the community please visit community.nginx.org.

For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

Step 14: Install PostgreSQL Client — apt install postgresql-client inside audit-tool

```
</body>  
</html>  
root@9eab4d718a89:/# apt install postgresql-client -y  
Installing:  
  postgresql-client
```

Step 15: psql Connection — psql -h postgres -U postgres connects successfully

```
Processing triggers for libc-bin (2.43-2ubuntu2) ...
root@9eab4d718a89:/# psql -h postgres -U postgres
Password for user postgres:
psql (18.4 (Ubuntu 18.4-0ubuntu0.26.04.1))
Type "help" for help.

postgres=# \l
```

Step 16: PostgreSQL II — List of databases: companydb, postgres, template0, template1

```
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ docker exec -it audit-tool bash
Setting up libgdbm6t64:amd64 (1.26-1build1) ...
Setting up libgdbm-compat4t64:amd64 (1.26-1build1) ...
Setting up libpq5:amd64 (18.4-0ubuntu0.26.04.1) ...
Setting up perl-modules-5.40 (5.40.1-7build1) ...
Setting up netbase (6.5build1) ...
Setting up readline-common (8.3-4) ...
Setting up libperl5.40:amd64 (5.40.1-7build1) ...
Setting up perl (5.40.1-7build1) ...
Setting up libreadline8t64:amd64 (8.3-4) ...
Setting up postgresql-client-common (290ubuntu1) ...
Setting up postgresql-client-18 (18.4-0ubuntu0.26.04.1) ...
update-alternatives: using /usr/share/postgresql/18/man/man1/psql.1.gz to provide /usr/share/man/man1/psql.1.gz (psql.1.gz) in auto mode
Setting up postgresql-client (18+290ubuntu1) ...
Processing triggers for libc-bin (2.43-2ubuntu2) ...
root@9eab4d718a89:/# psql -h postgres -U postgres
Password for user postgres:
psql (18.4 (Ubuntu 18.4-0ubuntu0.26.04.1))
Type "help" for help.

postgres=# \l

               List of databases
  Name  | Owner  | Encoding | Locale Provider | Collate  | Ctype    | Locale | ICU Rules | Access privileges
-----+-----+-----+-----+-----+-----+-----+-----+-----
companydb | postgres | UTF8     | libc             | en_US.utf8 | en_US.utf8 |         |           | 
postgres  | postgres | UTF8     | libc             | en_US.utf8 | en_US.utf8 |         |           | 
template0 | postgres | UTF8     | libc             | en_US.utf8 | en_US.utf8 |         |           | =c/postgres      +
          |          |          |          |          |          |         |           | postgres=CTc/postgres
template1 | postgres | UTF8     | libc             | en_US.utf8 | en_US.utf8 |         |           | =c/postgres      +
          |          |          |          |          |          |         |           | postgres=CTc/postgres
(4 rows)

postgres=#
```

Step 17: Audit Report Output — docker-audit-2026-06-17.txt shows images, networks, volumes, stats, ports

```
syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit/reports$ cat docker-audit-2026-06-17_10-31-58.txt

multistage-lab:latest      30371f3cdda3      92.7MB      26MB      U
nginx:latest               608a100c7165      241MB      66MB      U
postgres:latest           29ee7bb30d80      650MB      168MB      U
ubuntu:latest              f3d28607ddd7      160MB      45.3MB     U

==== Networks ====
NETWORK ID      NAME      DRIVER      SCOPE
b2d38f3b7c3c    bridge    bridge      local
383de4df043e    host      host        local
7772ea80cf7b    none      null        local
b3cce70deccf    security-net  bridge      local

==== Volumes ====
DRIVER      VOLUME NAME
local       audit-data
local       nginx-logs
local       postgres-data

==== Container Resource Usage ====
CONTAINER ID    NAME      CPU %      MEM USAGE / LIMIT      MEM %      NET I/O      BLOCK I/O      PIDS
4d67cdaa70f1    postgres  0.01%      32.41MiB / 7.387GiB     0.43%      4.68kB / 2.93kB  31.9MB / 54.8MB  9
36df70c7b877    nginx     0.00%      18.74MiB / 7.387GiB     0.25%      5.54kB / 5.27kB  16.9MB / 12.3kB  15
9eab4d718a89    audit-tool 0.00%      25.45MiB / 7.387GiB     0.34%      50.7MB / 1.39MB  175MB / 165MB    4

==== Port Mappings ====
NAMES      PORTS
postgres    0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
nginx       0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
audit-tool

==== Security-Net Details ====
[
0 0 0 0
```

Step 18: Final Project Tree — README.md, docker-compose.yml, audit.sh, postgres.env, reports/audit.txt

```
echo "Audit report saved to:"
echo "$REPORT_FILE"
● syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$ tree
.
├── README.md
├── docker-compose.yml
├── docker-security-audit.sh
├── postgres.env
├── reports
│   └── docker-audit-2026-06-17_10-31-58.txt
└──

2 directories, 5 files
○ syedaftab04@SYEDLENOVO:~/docker-labs/project-06-docker-security-audit$
```